Module 15) Advance Python Programming

7. Inheritance

- **Single, Multilevel, Multiple, Hierarchical, and Hybrid inheritance in Python**

Inheritance is a core concept of Object-Oriented Programming (OOP) where a class (called **child** or **derived class**) inherits attributes and methods from another class (called **parent** or **base class**). Python supports several types of inheritance:

1. Single Inheritance

- A child class inherits from **one** parent class.
- Simple one-to-one relationship.

Concept:
Child class extends or modifies behavior of a single parent.

2. Multilevel Inheritance

- Involves **multiple levels** of inheritance.
- A child class inherits from a parent class, which itself inherits from another parent.

Concept:
Grandparent → Parent → Child

3. Multiple Inheritance

- A child class inherits from **two or more** parent classes.

- Child class gets features from multiple classes.

4. Hierarchical Inheritance

- Multiple child classes inherit from **one** parent class.
- One base class → many derived classes.

5. Hybrid Inheritance

- Combination of two or more types of inheritance (multiple + multilevel).
- Complex inheritance structure mixing different patterns.

• Using the super() function to access properties of the parent class.

What is super()?

- super() is a built-in function used to **call methods or access properties from a parent (base) class**.
- It is commonly used in **inheritance** to **reuse or extend** functionality of the parent class.
- Helps avoid explicitly naming the parent class and makes code more maintainable.

Why Use super()?

- To **invoke the parent class's constructor (__init__)** or other methods.
- To **avoid duplicating code**.
- Supports **multiple inheritance** by properly resolving method calls.